

Show understanding of the need for: assembler software, a compiler, an interpreter

9608_s21_qp_11_sc_8.a.i

QUESTION

8 Jay is developing a computer game that allows users to create stories.

(a) Jay uses a language translator to develop the computer game.

(i) Tick (✓) **one or more** boxes in each row to identify the language translator(s) each statement describes.

Statement	Assembler	Interpreter	Compiler
Translates and executes each line of source code one line at a time			
Translates low-level source code into machine code			
Must be present in memory to execute the code			
Translates high-level source code into low-level code			

[4]

MARK SCHEME

8(a)(i)

1 mark per correct row

4

Statement	Assembler	Interpreter	Compiler
Translates and executes each line of source code one line at a time		✓	
Translates low-level source code into machine code	✓		
Must be present in memory to execute the code		✓	
Translates high-level source code into low-level code		✓	✓

9608_s20_qp_11_sc_8.b

QUESTION

- (b) The computer games are written in a high-level language. Bart does not need a compiler or an interpreter to run the games he buys for his console.

Explain why the games run without the need for a compiler or an interpreter.

.....
.....
.....
..... [2]

MARK SCHEME

8(b)	1 mark per bullet point to max 2 • Software will have been built using a compiler // the software is pre-compiled • Software is an executable file // the game is already in machine code // the game is already set-up to run on the console • Source code is not provided so does not need compiling/interpreting	2
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9608_w19_qp_11_sc_2.d

QUESTION

- (d) Aaron creates a web page using JavaScript code and HTML tags.

Describe how the JavaScript code is translated using an interpreter.

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..... [2]

MARK SCHEME

2(d)	1 mark per bullet point to max 2 The code is translated one line at a time ... and executed immediately The interpreter stops as soon as it finds an error	2
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9608_s19_qp_12_sc_2.a.i

QUESTION

2 Biyu is writing a computer program in a high-level language.

(a) Biyu uses a language translator.

(i) State the purpose of a language translator.

.....
..... [1]

MARK SCHEME

2(a)(i)	1 mark for correct answer To convert a (higher level) programming language to a different form	1
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9608_s19_qp_12_sc_2.a.iii

QUESTION

(iii) Name a translator other than an interpreter and a compiler.

..... [1]

MARK SCHEME

2(a)(iii)	1 mark for correct answer Assembler	1
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QUESTION

- (c) Kim develops her program and makes it ready for use. To do this, she uses first an interpreter and then a compiler.

Explain why Kim needs to use both an interpreter and a compiler.

Interpreter

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.....

.....

Compiler

.....

.....

.....

[4]

MARK SCHEME

6(c)	1 mark per bullet point. Max 3 marks for interpreter, max 3 marks for compiler Interpreter: • Used during development • Debugging is easier • ... Because errors are reported as they are found // No need to wait until the end of the process for the error report • ... Because errors can be corrected as they are found Compiler: • Compiler used when development complete // compiler used when program ready for distribution • Produces an executable file (.exe) • ... After compilation the compiler does not need to be present for the program to run • ... The program can be given to others without access to (source) code • ... Final program does not need to be re-compiled each time it is run • Cross-compilation, the program can be compiled to run on different platforms	4
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QUESTION

- 2 (a) The diagram shows three items of software that translate program code.

Draw **one** line from each context to the correct item of translation software.

Context	Item of translation software
A web page contains a client-side script.	Assembler
Each instruction in the source code consists of an op code and an operand.	Interpreter
The source code is required at run-time.	Compiler
When the source code is translated, copies of the executable program can be distributed without the need for the source code.	

[4]

MARK SCHEME

2(a)	The diagram shows four context boxes on the left and three software item boxes on the right. Arrows indicate the following connections: - From 'A web page contains a client-side script' to 'Interpreter'. - From 'Each instruction in the source code consists of an op code and an operand' to 'Interpreter'. - From 'The source code is required at run-time' to 'Compiler'. - From 'When the source code is translated, copies of the executable program can be distributed without the need for the source code' to 'Compiler'.	4
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QUESTION

- 2 (a) The diagram shows three items of software that translate program code.

Draw **one** line from each context to the correct item of translation software.

Context

The source code is written in a high-level language. An executable file is produced.

The source code uses instructions from the processor's instruction set.

The source code and translation software must both be in main memory at execution time.

A web page contains some JavaScript code.

Item of translation software

Assembler

Interpreter

Compiler

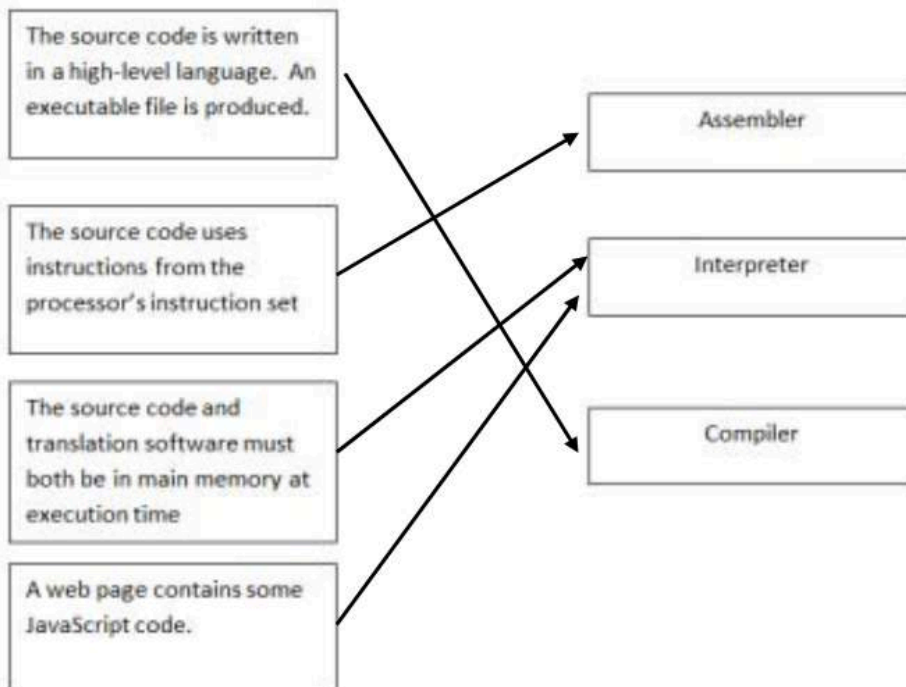
[4]

MARK SCHEME

2(a)

1 Mark for each correct connection

4



QUESTION

- 1 Three examples of language translators and four definitions are shown below.

Draw lines to link each language translator to the correct one or more definitions.

Language translator

Compiler

Assembler

Interpreter

Definition

The software reads the source code and reports all errors. The software produces an executable file.

The software reads each statement and checks it before running it. The software halts when it encounters a syntax error.

The software translates a high-level language program into machine code for the processor to execute.

The software translates low-level statements into machine code for the processor to execute.

[3]

MARK SCHEME

1 One mark for each box on the left.

[3]

Term

Definition

Compiler

The software reads the source code and reports all errors. The software produces an executable file.

Assembler

The software reads each statement and checks it before running it. The software halts when it encounters a syntax error.

Interpreter

The software translates a high level language program into machine code for the processor to execute

The software translates low-level statements into machine code for the processor to execute.

Marks allocated as follows:

- Compiler – 1 mark for **two** correct connecting lines
- Assembler – 1 mark for **one** correct connecting line
- Interpreter – 1 mark for **two** correct connecting lines

QUESTION

- 1 Three examples of language translators and four definitions are shown below.

Draw lines to link each language translator to the correct one or more definitions.

Language translator

Compiler

Assembler

Interpreter

Definition

The software reads the source code and reports all errors. The software produces an executable file.

The software reads each statement and checks it before running it. The software halts when it encounters a syntax error.

The software translates a high-level language program into machine code for the processor to execute.

The software translates low-level statements into machine code for the processor to execute.

[3]

MARK SCHEME

1 One mark for each box on the left.

[3]

Term

Definition

Compiler

The software reads the source code and reports all errors. The software produces an executable file.

Assembler

The software reads each statement and checks it before running it. The software halts when it encounters a syntax error.

Interpreter

The software translates a high level language program into machine code for the processor to execute

The software translates low-level statements into machine code for the processor to execute.

Marks allocated as follows:

- Compiler – 1 mark for **two** correct connecting lines
- Assembler – 1 mark for **one** correct connecting line
- Interpreter – 1 mark for **two** correct connecting lines

QUESTION

- 11 A game program is written which can be either interpreted or compiled. The table below shows five statements about the use of interpreters and compilers.

Tick (✓) to show whether the statement refers to an interpreter or to a compiler.

Statement	Interpreter	Compiler
This translator creates an executable file		
When this translator encounters a syntax error, game execution halts		
The translator analyses and checks each line just before executing it		
This translator will produce faster execution of the game program		
Use of this translator makes it more difficult for the user to modify the code of the game		

[5]

MARK SCHEME

11

Statement	Interpreter	Compiler
This translator creates an executable file		✓
When this translator encounters a syntax error, game execution will halt	✓	
The translator analyses and checks each line just before executing it	✓	
This translator will produce faster execution of the game program		✓
Use of this translator makes it more difficult for the user to modify the code of the game supplied to the user		✓

1 mark for each correct row

[5]

QUESTION

- 11 A game program is written which can be either interpreted or compiled. The table below shows five statements about the use of interpreters and compilers.

Tick (✓) to show whether the statement refers to an interpreter or to a compiler.

Statement	Interpreter	Compiler
This translator creates an executable file		
When this translator encounters a syntax error, game execution halts		
The translator analyses and checks each line just before executing it		
This translator will produce faster execution of the game program		
Use of this translator makes it more difficult for the user to modify the code of the game		

[5]

MARK SCHEME

11

Statement	Interpreter	Compiler
This translator creates an executable file		✓
When this translator encounters a syntax error, game execution will halt	✓	
The translator analyses and checks each line just before executing it	✓	
This translator will produce faster execution of the game program		✓
Use of this translator makes it more difficult for the user to modify the code of the game supplied to the user		✓

1 mark for each correct row

[5]

Explain the benefits and drawbacks of using either a compiler or interpreter and justify the use of each

9608_w21_qp_11_sc_10.a

QUESTION

10 Janaka is developing a new computer program. She decides to use an interpreter instead of a compiler.

(a) State **three** benefits of using an interpreter.

- 1
-
- 2
-
- 3
-

[3]

MARK SCHEME

10(a)	One mark per bullet point to max 3 • Easier de-bugging • Errors can be corrected in real time • Errors are reported as the interpreter finds them. • An error can be corrected and translation continued from where it stopped • The effect of any change made to the code can be seen immediately • Parts of the program can be tested, without all the program code being available.	3
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9608_w21_qp_11_sc_10.b

QUESTION

(b) State **one** drawback of using an interpreter.

-
- [1]

MARK SCHEME

10(b)	One mark for a correct answer Source code is needed at run time // No executable file produced, (so source code can be edited) // Translation software needed every time the program is run // execution time increased	1
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9608_s21_qp_13_sc_7.a

QUESTION

7 Patrick is writing a new software application. He is using a compiler to develop the software application.

(a) Describe the drawbacks of using a compiler instead of an interpreter.

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.....

.....

.....

.....

..... [3]

MARK SCHEME

7(a)	1 mark per bullet point to max 3 • The program will not run if there are any errors • The program must be recompiled after every change // cannot correct errors in real-time • Part-programs cannot be tested • One error may result in other false errors being reported	3
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QUESTION

8 Jay is developing a computer game that allows users to create stories.

(a) Jay uses a language translator to develop the computer game.

(i) Tick (✓) **one or more** boxes in each row to identify the language translator(s) each statement describes.

Statement	Assembler	Interpreter	Compiler
Translates and executes each line of source code one line at a time			
Translates low-level source code into machine code			
Must be present in memory to execute the code			
Translates high-level source code into low-level code			

[4]

MARK SCHEME

8(a)(i)

1 mark per correct row

4

Statement	Assembler	Interpreter	Compiler
Translates and executes each line of source code one line at a time		✓	
Translates low-level source code into machine code	✓		
Must be present in memory to execute the code		✓	
Translates high-level source code into low-level code		✓	✓

9608_s21_qp_11_sc_8.a.ii

QUESTION

- (ii) Jay decides to use a compiler to develop the game.

Identify **two** benefits of using a compiler.

Benefit 1

.....

Benefit 2

.....

[2]

MARK SCHEME

8(a)(ii)	1 Mark per bullet point to max 2 • The compiler is not required to run the program • The program can be distributed without the source code • A compiler produces code that executes faster than the equivalent for an interpreter	2
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9608_s20_qp_13_sc_4.b

QUESTION

- (b) Each member of the group is creating a different part of the game. Each person needs to test their part of the game independently before they are combined.

Identify the **most appropriate** type of translator that should be used to test each part of the game independently. Justify your choice.

Translator

Justification

.....

.....

.....

.....

[3]

MARK SCHEME

4(b)	1 mark for interpreter 1 mark per bullet point for justification to max 2 • The interpreter translates and executes line by line • The interpreter stops translation when an error is found • ...the error can be corrected, and the program continues running from where it stopped • There is no need for an executable file (at this stage)	3
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9608_s20_qp_11_sc_8.b

QUESTION

(b) The computer games are written in a high-level language. Bart does not need a compiler or an interpreter to run the games he buys for his console.

Explain why the games run without the need for a compiler or an interpreter.

.....

.....

.....

..... [2]

MARK SCHEME

8(b)	1 mark per bullet point to max 2 • Software will have been built using a compiler // the software is pre-compiled • Software is an executable file // the game is already in machine code // the game is already set-up to run on the console • Source code is not provided so does not need compiling/interpreting	2
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9608_w19_qp_12_sc_5.b

QUESTION

(b) Identify **one** way Mica can distribute the software without the source code.

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..... [1]

MARK SCHEME

5(b)	Provide an .exe file // Compile the source code // Use a compiler	1
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QUESTION

(e) When the program is complete, Frankie uses a compiler to prepare the program for the client.

Explain why Frankie uses a compiler instead of an interpreter.

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..... [3]

MARK SCHEME

2(e)	1 mark per bullet point to max 3 He uses a compiler because it creates a separate executable file The executable means the client cannot access the source code // edit the program The executable means the client does not need the IDE / compiler It will (probably) be faster to run the executable than to interpret the source code every time the program is run	3
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QUESTION

(ii) Biyu uses an interpreter.

State **two** benefits of Biyu using an interpreter instead of a compiler while writing the program.

1

.....

2

.....

[2]

MARK SCHEME

2(a)(ii)	1 mark per bullet point to max 2 - Easier de-bugging ...because errors can be corrected in real time - The effect of any change made to the code can be seen immediately - Parts of the program can be <u>tested</u>, without all the program code being available	2
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9608_s19_qp_11_sc_3.b.i

QUESTION

(b) Three translators are compilers, interpreters, and assemblers.

(i) State **one** benefit of Kimmy using an **interpreter** during the development of the program.

.....

..... [1]

MARK SCHEME

3(b)(i)	1 mark per bullet point to max 1 Errors can be corrected as they occur Can run a partially complete program when developing The effect of any change made to the code can be seen immediately	1
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9608_s19_qp_11_sc_3.b.ii

QUESTION

(ii) State **three** benefits of Kimmy using a **compiler** when the program is complete.

1

.....

2

.....

3

..... [3]

MARK SCHEME

3(b)(ii)	1 mark per bullet point to max 3 Produces an executable file User does not have access to source code It will (probably) be faster to run the executable Code does not have to be compiled each time it is run Does not need the compiler to be present at run-time	3
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9608_w18_qp_12_sc_6.c

QUESTION

- (c) Kim develops her program and makes it ready for use. To do this, she uses first an interpreter and then a compiler.

Explain why Kim needs to use both an interpreter and a compiler.

Interpreter

.....

.....

.....

Compiler

.....

.....

.....

[4]

MARK SCHEME

6(c)	1 mark per bullet point. Max 3 marks for interpreter, max 3 marks for compiler Interpreter: • Used during development • Debugging is easier • ... Because errors are reported as they are found // No need to wait until the end of the process for the error report • ... Because errors can be corrected as they are found Compiler: • Compiler used when development complete // compiler used when program ready for distribution • Produces an executable file (.exe) • ... After compilation the compiler does not need to be present for the program to run • ... The program can be given to others without access to (source) code • ... Final program does not need to be re-compiled each time it is run • Cross-compilation, the program can be compiled to run on different platforms	4
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9608_w17_qp_13_sc_2.a

QUESTION

2 (a) The diagram shows three items of software that translate program code.

Draw **one** line from each context to the correct item of translation software.

Context

A web page contains a client-side script.

Each instruction in the source code consists of an op code and an operand.

The source code is required at run-time.

When the source code is translated, copies of the executable program can be distributed without the need for the source code.

Item of translation software

Assembler

Interpreter

Compiler

[4]

MARK SCHEME

2(a)	<pre> graph LR A[A web page contains a client-side script] --> Assembler B[Each instruction in the source code consists of an op code and an operand] --> Assembler C[The source code is required at run-time] --> Interpreter D[When the source code is translated, copies of the executable program can be distributed without the need for the source code] --> Compiler </pre>	4
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9608_w17_qp_12_sc_2.b.i

QUESTION

(b) A programmer is developing software and has both a compiler and interpreter for the high-level language used. Describe two benefits of using each form of translation software. (i) Benefits of a compiler 1 2 (ii) Benefits of an interpreter [2]

MARK SCHEME

2(b)(i)	1 Mark per bullet, max 2 Once translated the compiler software is not needed to run the program Compiled code should execute faster Compiler produces an executable file The executable file produced by a compiler can be distributed without users having sight of the source code // source code is kept secure // users are unable to make changes to the program Cross-compilation is possible	2
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QUESTION

(ii) Benefits of an interpreter

- 1
-
- 2
-

[2]

MARK SCHEME

2(b)(ii)	1 Mark per bullet, max 2 Easier de-bugging The interpreter stops when error encountered error can be corrected in real time The interpreter translates a statement then executes it immediately Parts of the program can be tested, without all the program code being available.	2
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QUESTION

1 Describe **two** differences between a compiler and interpreter.

- 1
-
-
-
- 2
-
-
-

[4]

MARK SCHEME

1 Four from:

[4]

- Compiler creates an executable//an interpreter does not create an executable.
- The compiled program can be independently distributed.
- Compiler reports all errors at the end of compilation//an interpreter stops when it reaches an error.
- Interpreter executes each statement immediately after decoding/checking it//a compiler checks the whole program for errors.
- The interpreter software/source code must be present in main memory every time the program is executed//the compiled program does not require compiler/source code to be present.
- Cross-compilation is possible/compile on one hardware platform to run on another.

9608_w15_qp_13_sc_11

QUESTION

- 11 A game program is written which can be either interpreted or compiled. The table below shows five statements about the use of interpreters and compilers.

Tick (✓) to show whether the statement refers to an interpreter or to a compiler.

Statement	Interpreter	Compiler
This translator creates an executable file		
When this translator encounters a syntax error, game execution halts		
The translator analyses and checks each line just before executing it		
This translator will produce faster execution of the game program		
Use of this translator makes it more difficult for the user to modify the code of the game		

[5]

MARK SCHEME

11

Statement	Interpreter	Compiler
This translator creates an executable file		✓
When this translator encounters a syntax error, game execution will halt	✓	
The translator analyses and checks each line just before executing it	✓	
This translator will produce faster execution of the game program		✓
Use of this translator makes it more difficult for the user to modify the code of the game supplied to the user		✓

1 mark for each correct row

[5]

9608_w15_qp_11_sc_11

QUESTION

- 11 A game program is written which can be either interpreted or compiled. The table below shows five statements about the use of interpreters and compilers.

Tick (✓) to show whether the statement refers to an interpreter or to a compiler.

Statement	Interpreter	Compiler
This translator creates an executable file		
When this translator encounters a syntax error, game execution halts		
The translator analyses and checks each line just before executing it		
This translator will produce faster execution of the game program		
Use of this translator makes it more difficult for the user to modify the code of the game		

[5]

MARK SCHEME

11

Statement	Interpreter	Compiler
This translator creates an executable file		✓
When this translator encounters a syntax error, game execution will halt	✓	
The translator analyses and checks each line just before executing it	✓	
This translator will produce faster execution of the game program		✓
Use of this translator makes it more difficult for the user to modify the code of the game supplied to the user		✓

1 mark for each correct row

[5]

Show awareness that high-level language programs may be partially compiled and partially interpreted, such as Java (console mode)

9608_w17_qp_13_sc_2.b.i

QUESTION

(b) The Java programming language is said to be machine or platform independent.

(i) Describe what is meant by **machine independent**.

.....
.....[1]

MARK SCHEME

2(b)(i)	One mark from: The program code can be translated to run on any processor / platform Source code is translated into machine independent intermediate code not machine dependent code	1
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9608_w17_qp_13_sc_2.b.ii

QUESTION

(ii) Describe how a Java source code program is translated.

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.....
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.....[2]

MARK SCHEME

2(b)(ii)	Two marks from: Java uses a two-step translation process Java code is partially interpreted – partially compiled Code is translated first into intermediate code / "bytecode"... ...using the Java compiler The bytecode is finally interpreted by the Java Virtual Machine	Max 2
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Describe features found in a typical Integrated Development Environment (IDE)
